

Hierarchical optimize scheduling strategy for orderly charging of electric vehicles based on PTM and CNN-Transformer-LightGBM model[☆]

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ABSTRACT

With the significant increase in the penetration rate of electric vehicles, uncoordinated charging of EV clusters can exacerbate peak-to-valley differences, resulting in a "peak on peak" phenomenon. This paper proposes a hierarchical optimize scheduling strategy for orderly charging based on the Probability Transition Matrix algorithm (PTM) using CNN-Transformer-LightGBM. Firstly, we establish electric vehicle load prediction models for CNN-Transformer-LightGBM separately, and use the inverse variance method to weight and combine the two models into a CNN-Transformer-LightGBM composite model; To optimize the continuous parameters within the model, TPM is used for hyperparameter optimization to achieve optimal charging control. Simulation results indicate that the proposed strategy effectively reduces the grid load's peak-to-valley difference by 43 % and decreases comprehensive grid load variance by 32 %.

1. Introduction

Transportation electrification is one of the key approaches to achieving carbon neutrality goals. Electric vehicles (EVs) use electricity to replace traditional carbon-based fossil fuels, thereby significantly reducing fossil fuel consumption and greenhouse gas emissions. However, EV charging behavior is characterized by randomness and uncertainty [1,2]. When EVs are integrated into the grid at high penetration rates, large-scale uncoordinated EV charging and discharging behavior can pose a significant threat to grid stability [3,4]. At the same time, with the rapid development of renewable energy technologies, the installed capacity of wind and photovoltaic power generation continues to grow, increasing the intermittency of renewable energy and the uncertainty of EV charging behavior, which makes grid management more challenging.

The widespread adoption of EVs will also lead to increased electricity consumption, adding additional pressure on grid operation. EV charging events are related to user driving habits, and the charging load has stochastic uncertainty. Uncoordinated EV charging behavior endangers grid safety as well as the healthy development of EVs. Therefore, it is necessary to formulate appropriate measures to manage EV charging

behavior. If a large number of EVs charge during peak periods, grid stability will be affected, leading to increased line losses, reduced power quality, and shortened transformer life. Literature [5], a load forecasting model based on Bi-LSTM and feature correlation analysis was established, which improved prediction accuracy. Literature [6] proposed a method combining modal decomposition and attention mechanisms. Literature [7] introduced a forecasting method that integrates load decomposition and Long Short-Term Memory (LSTM) networks, demonstrating that extracting data feature components can effectively enhance accuracy. Literature [8] developed a short-term load forecasting method for electric vehicle charging stations based on an improved optimization algorithm combined with feature extraction and Gated Recurrent Unit (GRU). Literature [9] constructed an ultra-short-term load forecasting model for electric vehicle charging stations using a combination of ensemble learning and the Light Gradient Boosting Machine (LightGBM) framework. Literature [10] proposed a method that first applies K-means clustering analysis before performing load forecasting. Literature [11] suggested a combination of spectral clustering and LSTM for predicting the charging load of electric buses, achieving a significant improvement in prediction accuracy. However, the aforementioned methods are prone to oscillations in peak

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and trough predictions, failing to meet higher accuracy requirements. In contrast, algorithms based on the Transformer architecture demonstrate superior precision in time series load forecasting while enabling better model parallelism. Reference [12] developed a system to optimize charging coordination, which not only meets EV energy needs but also improves the load curve. Based on the mixed-integer linear programming framework, demand response charging performed better than other charging schemes in terms of improving voltage, reducing peak load, and reducing net-work losses. Reference [13] designed an incentive-based demand response mechanism to balance the interests of grid operators, EV aggregators, and EV customers in the distribution network. Studies have shown that by modifying user response rates, adjustable benchmark compensation prices can mitigate insufficient response, reduce economic costs, and reduce peak loads. Reference [14] proposed a stochastic two-stage optimization model for EV participation in demand response, where the upper objective is to maximize the profits of EV aggregators, and the lower objective is to minimize EV charging costs. Existing studies have explored various approaches for EV charging load forecasting. Traditional methods, such as Bi-LSTM [7], GRU [8], and LightGBM [9], leverage feature correlation analysis and ensemble learning to improve accuracy. Advanced techniques like spectral clustering [10] and modal decomposition [11] address temporal variability but struggle with peak-trough oscillation and long-term dependency modeling. Transformer-based algorithms [12–14] demonstrate superior performance in time-series forecasting due to their parallel computation and self-attention mechanisms, yet their integration with spatial feature extraction remains underexplored.

Reference [15] proposed a real-time pricing method for controlling plug-in hybrid EV charging, evaluating the impact of energy prices and charging willingness on charging load. Distributed algorithms were used to calculate appropriate real-time charging prices to achieve load shifting and energy management. Reference [16] proposed an energy management strategy that considers solar renewable energy, market electricity prices, and EV uncertainty, improving the profitability of smart industrial grids through demand response and EV optimization. Reference [17] regarded EVs as demand-side response resources under dynamic pricing. The developed charging method aimed to minimize the unpredictability of renewable energy generation. The optimized genetic algorithm effectively reduced the volatility of renewable energy generation and EV charging costs. Reference [18] scientifically divided load periods based on price-based demand response and formulated corresponding electricity prices for each period. Existing research has made significant contributions to optimizing EV charging load through price incentives. However, detailed studies on load cycle division are often based on empirical applications, which can lead to overlap between new energy and traditional load cycles. Although strategies such as demand response, charging optimization, and dynamic pricing can regulate EV charging behavior to some extent, they still face challenges in prediction accuracy, making it difficult to avoid "peak on peak" issues.

While our proposed hierarchical scheduling strategy leverages the strengths of CNN-Transformer and LightGBM to mitigate these gaps, it is important to acknowledge potential limitations inherent in this initial investigation. The study relies on simulated datasets for EV behavior and renewable generation modeling, which may affect the generalizability of results to real-world grid environments with varying configurations, penetration levels, or extreme scenarios such as blackouts or highly volatile renewable output. Additionally, the Gaussian Mixture Model (GMM) assumptions for characterizing EV arrival times and SOC, while effective in capturing multi-peak distributions, may not fully account for region-specific user profiles.

To address these challenges, this paper addresses the issue of increased peak-to-valley differences and the "peak on peak" phenomenon caused by uncoordinated charging of electric vehicle (EV) clusters by proposing an orderly charging scheduling strategy based on the TPM algorithm and the CNN-Transformer-LightGBM model. The strategy

integrates the CNN-Transformer and LightGBM models using the inverse variance weighting method to predict the charging power guidance curve for EV clusters. Additionally, an improved probabilistic transition matrix (TPM) algorithm is employed to achieve hierarchical control objectives, including minimizing the variance of the comprehensive grid load (incorporating wind and photovoltaic power generation) and mitigating grid load fluctuations caused by uncoordinated EV charging. This work specifically pursues three primary objectives: (1) establishing an accurate charging load prediction framework robust to renewable intermittency through hybrid CNN-Transformer-LightGBM modeling, (2) developing hierarchical control via probabilistic transition matrix optimization for real-time scheduling, and (3) enabling scalable deployment through lightweight algorithms requiring no grid hardware upgrades. This approach enables global optimization and local fine-grained control of charging loads. Simulation results show that the main control system effectively coordinates the collaborative operation of various energy sources and loads in the grid, optimizing power resource allocation. The secondary control system arranges charging periods flexibly based on the main control system's guidance curve, ensuring that EV charging needs are met while keeping the deviation between actual charging power and the guidance power within an acceptable range. This strategy effectively improves the stability of grid operation and ensures efficient management of EV charging.

2. EV user characterization analysis

The charging power of individual EVs connected to the grid is determined by the charging pile, but the grid-connected power of EVs is often influenced by user characteristics. Methods for characterizing EV user characteristics typically include regression analysis, clustering analysis, decision trees, and machine learning algorithms. Current research primarily uses normal probability distribution fitting since EV user travel patterns have morning and evening peaks, and a GMM can accurately describe these multi-peak distributions.

Therefore, this paper uses GMM to classify and fit the probability density function (PDF) established based on actual data for EV arrival times ($T_{\text{arrive},i}$) and initial states of charge (State of Charge, SOC). The specific probability distribution function ($P(T_{\text{arrive},i}|\Theta)$) for the arrival time of the m^{th} EV at the charging pile is given as follows:

$$P(T_{\text{arrive},i}|\Theta) = \sum_{j=1}^{\theta} \alpha_j N_{\theta}(T_{\text{arrive},i}; \mu_{\theta}, D_{\theta}) \quad (1)$$

$$N_{\theta}(T_{\text{arrive},i}; \mu_{\theta}, D_{\theta}) = \frac{1}{\sqrt{2\pi D_{\theta}}} \exp\left[-\frac{1}{2D_{\theta}}(T_{\text{arrive},i} - \mu_{\theta})^2\right] \quad (2)$$

Where $N_{\theta}(T_{\text{arrive},i}; \mu_{\theta}, D_{\theta})$ represents the θ^{th} Gaussian component function, μ_{θ} is the mean, and D_{θ} is the variance. J is the total number of Gaussian components, and α_{θ} is the weight of the Gaussian component. The one-dimensional GMM is expressed as $\Theta = (\theta_1, \dots, \theta_J)$. By determining $T_{\text{arrive},i}$, μ_{θ} , D_{θ} , the Gaussian mixture distribution is fitted, and J is adjusted based on the actual fitting effect after estimation.

The initial energy probability density function $P_d(E_{m0}; \varphi_m, D_m)$ of the m^{th} EV is expressed as:

$$P_d(E_{m0}; \varphi_m, D_m) = \sum_{j=1}^{m-1} \frac{\tau_m}{d_{mR}(1 - \text{SOC}_{m0}\sqrt{2\pi D_m})} \times \exp\left(1 - (\ln(1 - \text{SOC}_{m0}) - (\varphi_m - \ln(d_{mR}/\tau_m)))^2 / (2D_m)\right) \quad (3)$$

Where SOC_{m0} is the initial state of charge of the m^{th} EV, φ_m is the daily driving mileage, D_m is the variance of the probability density function, and τ_m is the number of days the EV can drive after charging. d_{mR} represents the EV's driving range. E_{m0} is the initial energy of the m^{th} EV.

The parameters of the initial SOC probability density function are estimated based on historical data, and the fitted density functions for

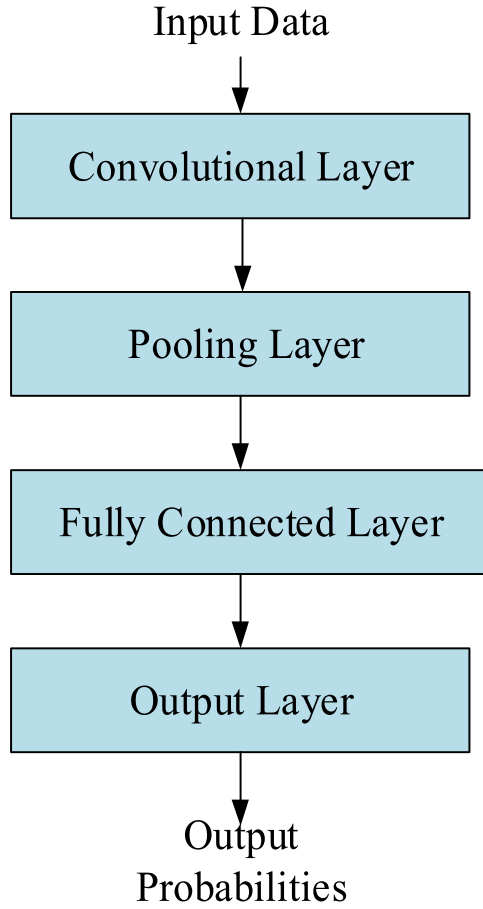


Fig. 1. CNN structure.

SOC_0 and T_{arrive} are derived. During the simulation, random numbers are drawn from the respective probability densities using the Monte Carlo method. Based on user-set stopping durations, the departure time $T_{deprive}$ is obtained for each EV (a total of EV_{Number}) to characterize user behavior.

3. Electric vehicle load forecasting model based on the CNN-transformer-LightGBM hybrid algorithm

Inspired by the ability of the LightGBM algorithm to capture relationships within data and the advantages of the Transformer model, a LightGBM-Transformer hybrid model is proposed to improve short-term power load forecasting accuracy. This model leverages both load data and time/weather features. First, the LightGBM algorithm is used to perform feature selection on the multi-feature input, removing redundant features. Then, a sliding time window is constructed, and the selected features and load data are used as new input for the Transformer model to predict electric vehicle load. This approach combines the strengths of both models, enhancing prediction performance.

3.1. CNN-transformer prediction model

1) CNN Model

As shown in Fig. 1, the structure of a CNN primarily consists of an input layer, convolutional layer, pooling layer, fully connected layer, and output layer. First, the dataset is fed into the model through the input layer. Next, the convolutional layer and pooling layer work together to extract and filter features from the input data. This process not only captures key features but also significantly reduces

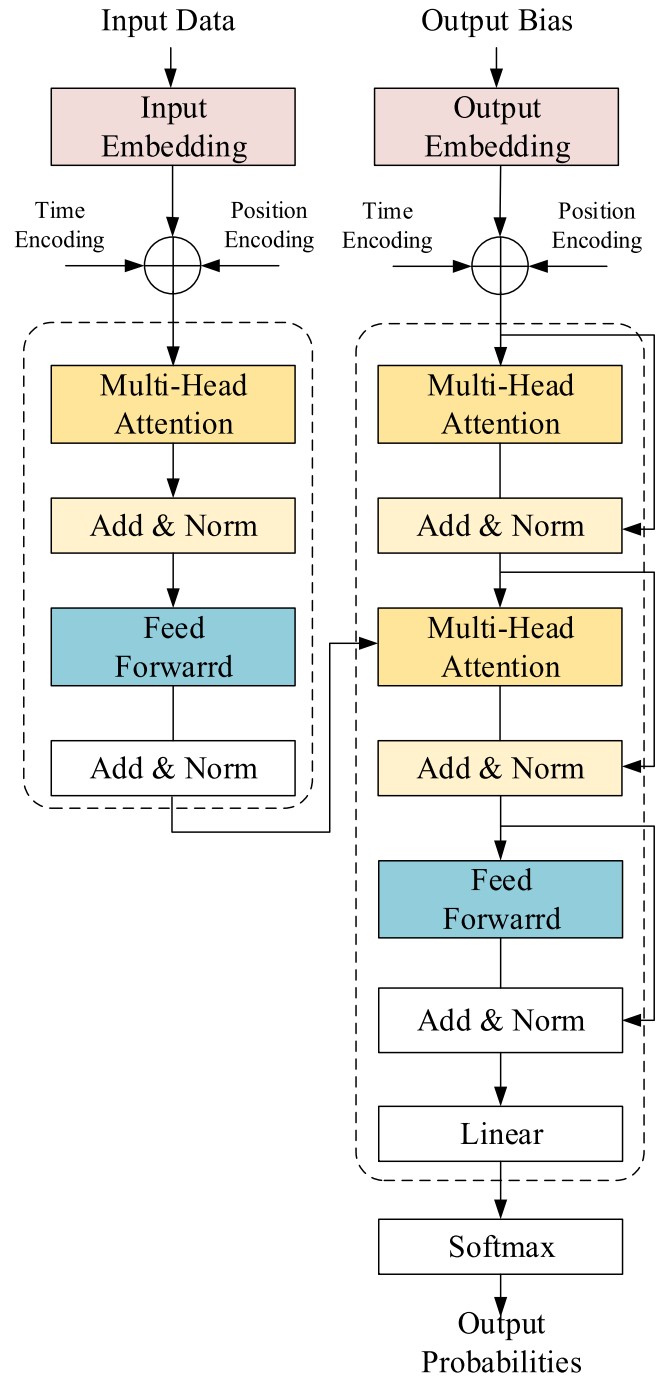


Fig. 2. Transformer Model Structure.

computational complexity, improving the model's operational efficiency [19]. Subsequently, the extracted features are passed to the fully connected layer, whose main function is to integrate feature information from different levels, further mining deep-level feature representations within the data. Through this mechanism, the model can fully learn the underlying patterns in the data. After the fully connected layer completes feature integration, the data is finally passed to the output layer. The output layer maps the integrated features into the model's final prediction results. This hierarchical structural design enables the CNN to efficiently capture the inherent patterns in the data, thereby achieving stable and accurate predictive performance.

2) Transformer Model

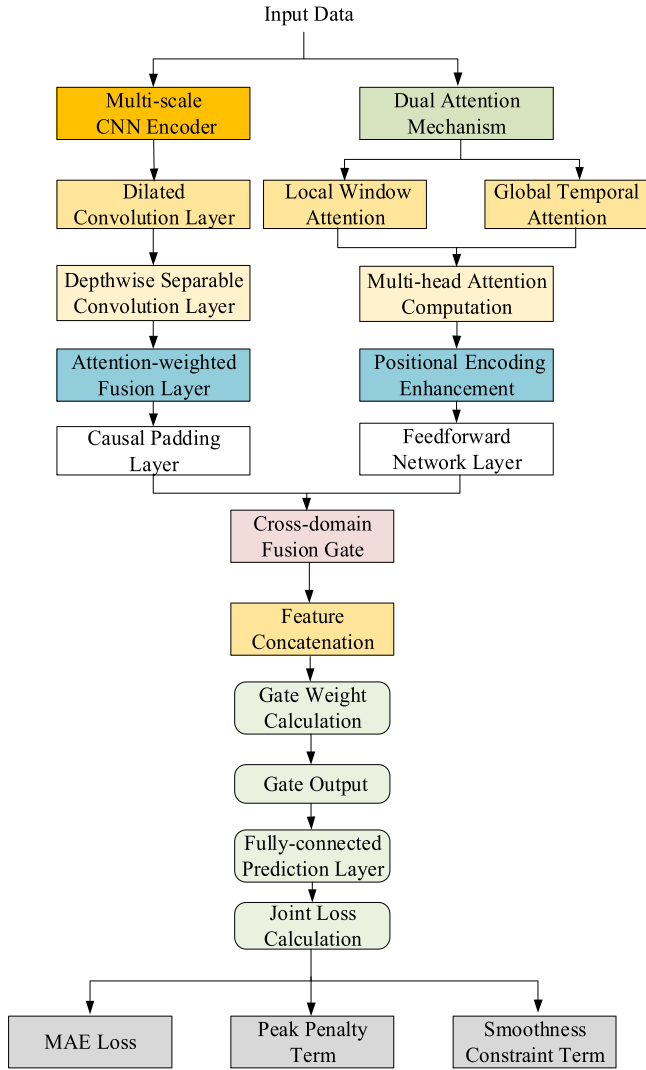


Fig. 3. Enhanced CNN-Transformer Model Structure.

The original Transformer model, based on the self-attention mechanism, was initially applied in the field of Natural Language Processing (NLP). Due to its powerful capabilities, it has since been widely used in various fields. The Transformer employs an encoder-decoder structure to process data, and its architecture is shown in Fig. 2. The Transformer is divided into four main components: the input data and target values, position encoding, the encoder-decoder structure, and the output layer.

The encoder consists of N identical stacks, with each layer composed of two sub-layers: a multi-head self-attention mechanism and a feed-forward neural network. A residual connection is used around each sub-layer, followed by layer normalization. The depth of all modules in the model is unified to the same model depth d_{model} for parameter consistency. The decoder consists of N identical stacks, with an additional third sub-layer that performs multi-head attention on the encoder's output. To ensure that the prediction for position i only depends on known outputs for positions less than i , the de-coder's self-attention sub-layer is masked.

The output layer decodes the data transmitted from the decoder using a fully connected layer and the softmax function. During prediction tasks, the target values are null. User behavior data is generated using GMM fitted to historical data, ensuring realistic charging behavior diversity.

3) The Enhanced CNN-Transformer Model

This paper proposes an enhanced CNN-Transformer model, primarily designed for electric vehicle load forecasting tasks. The process begins with "input data" and passes through multiple processing layers and modules, ultimately reaching outputs such as "MAE loss," "peak penalty," and "smoothness constraint." First, the input data undergoes feature extraction through the "multi-scale CNN encoder," which utilizes "dilated convolution layers" and "depth wise separable convolution layers" to capture features at different time scales. Next, the extracted features are fed into the "dual-attention Transformer" module, which captures local dependencies within short time windows and global temporal dependencies through "local window attention" and "global temporal attention," respectively. The feature representation is further optimized using "multi-head attention computation," "attention-weighted fusion layer," "positional encoding enhancement," "causal padding layer," and "feedforward network layer." Subsequently, the features extracted by CNN and Transformer are dynamically fused through the "cross-domain fusion gate" module, specifically including "feature concatenation," "gating weight computation," and "gated output." The fused features are then passed to the "fully connected prediction layer" to generate the final prediction results. Finally, the model's training objective is achieved through "joint loss computation," which includes "MAE loss" to measure the absolute error between predicted and true values, "peak penalty" to ensure accurate prediction of peak loads, and "smoothness constraint" to suppress abrupt changes in predicted values between adjacent time steps, thereby generating smooth predictions.

The multi-scale CNN encoder captures features at different time scales using dilated convolutions. The dilation rate increases exponentially ($d=1$ for a dilation rate of 1, $d=3$ for a dilation rate of 4), enabling the model to extract features ranging from minute-level to weekly-level. Specifically, the output feature of the d -th dilated convolution branch at the l -th layer can be expressed as:

$$\mathbf{H}_d^{(l)} = \text{Conv}(\mathbf{X}, \mathbf{W}_d, r_d) \quad (4)$$

Where $\mathbf{W}_d$ represents the learnable convolution kernel parameters, and r_d is the dilation rate. To fuse features from different scales, an attention-weighted mechanism is introduced, where learnable attention weight α_d generated by Softmax are used to weight features across different time scales (hour/day/week):

$$\mathbf{H}_{\text{fusion}} = \sum_{d=1}^3 \alpha_d \cdot \mathbf{H}_d^{(l)}, \quad \sum \alpha_d = 1 \quad (5)$$

The dual-attention Transformer module captures both local and global dependencies in the time series through local window attention and global temporal attention. The local window attention uses a 24-hour window with a banded mask $\mathbf{M}_{\text{local}}$, while the global temporal attention reduces computational complexity by down sampling the key matrix $\mathbf{K}_{\text{global}}$. Specifically, the local attention $\text{Attn}_{\text{local}}$ and global attention $\text{Attn}_{\text{global}}$ are calculated as follows:

$$\text{Attn}_{\text{local}} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{W}_Q(\mathbf{K}\mathbf{W}_K)^\top}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V}\mathbf{W}_V \quad (6)$$

$$\text{Attn}_{\text{global}} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{W}_Q(\mathbf{K}_i\mathbf{W}_K)^\top}{\sqrt{d_k}}\right)\mathbf{V}\mathbf{W}_V \quad (7)$$

Here, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ represent the query, key, and value matrices, respectively, and d_k is the dimensionality of each attention head. Finally, the results of local and global attention are concatenated and normalized using LayerNorm:

$$\mathbf{H}_{\text{trans}} = \text{LayerNorm}(\text{Concat}(\text{Attn}_{\text{local}}, \text{Attn}_{\text{global}})) \quad (8)$$

To further optimize model performance, a cross-domain fusion gating mechanism is introduced. This mechanism dynamically adjusts the contributions of CNN and Transformer using a gating weight g_i .

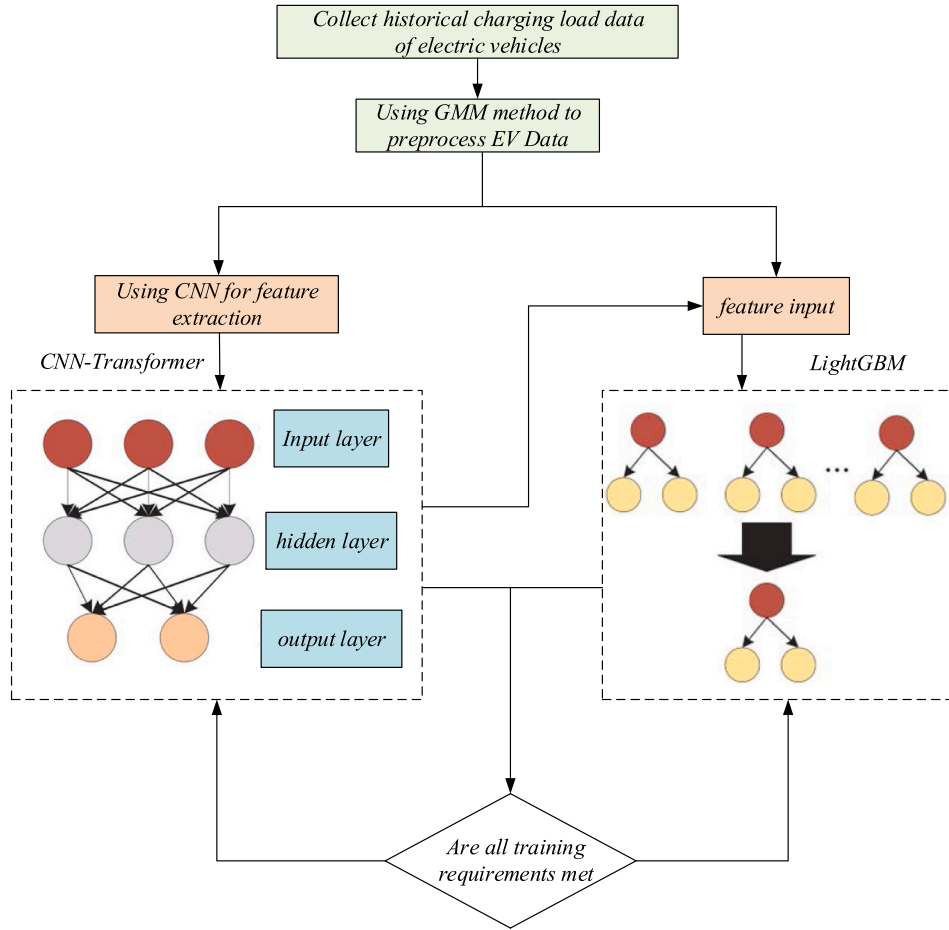


Fig. 4. EV Load Forecasting Process.

Specifically, the gating value g_t is generated by the Sigmoid activation function and dynamically adjusted based on the historical average load baseline $\bar{y}_t$ and the allowed peak fluctuation threshold (e.g., 1.2). The feature integration formula is as follows:

$$g_t = \sigma(\mathbf{W}_g [\text{AvgPool}(\mathbf{H}_{\text{conv}}) \parallel \text{MaxPool}(\mathbf{H}_{\text{trans}})] + \mathbf{b}_g) \quad (9)$$

$$\mathbf{H}_{\text{final}} = g_t \odot \mathbf{H}_{\text{conv}} + (1 - g_t) \odot \mathbf{H}_{\text{trans}} \quad (10)$$

To suppress abrupt changes in predicted values between adjacent time steps and meet the ramp rate constraints in grid scheduling, a smoothness constraint is introduced. The loss function for the smoothness constraint is defined as:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{T} \sum_{t=1}^T |y_t - \hat{y}_t| \quad (11)$$

$$\mathcal{L}_{\text{peak}} = \frac{1}{T} \sum_{t=1}^T \max(0, \hat{y}_t - 1.2\bar{y})^2 \quad (12)$$

$$\mathcal{L}_{\text{smooth}} = \frac{1}{T-1} \sum_{t=2}^T (\hat{y}_t - \hat{y}_{t-1})^2 \quad (13)$$

In terms of computational complexity, the ratio of local attention to global attention complexity is $L=96$. When the sequence length $L=168$, the computational cost is reduced by 98.25 % compared to weighted attention. This design significantly reduces the computational burden while maintaining high prediction accuracy.

3.2. LightGBM model

In the LightGBM model, the optimization process of the objective function is simplified through the Taylor expansion of the loss function [20]. Specifically, the objective function consists of the loss function and a regularization term, which are used to measure the prediction error and complexity of the model. The first-order gradient g_i and the second-order gradient h_i of the loss function represent the first and second derivatives of the loss function for sample i with respect to the predicted value $\hat{y}_i$. The total objective function L of the model can be expressed as:

$$\mathcal{L}(\theta) = \sum_{i=1}^n L(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (14)$$

Where $L(y_i, \hat{y}_i)$ is the loss function for sample i , measuring the difference between the true value y_i and the predicted value $\hat{y}_i$; $\Omega(f_k)$ is the complexity regularization term for the k -th tree, used to prevent overfitting.

During the tree generation process, the calculation of the split gain is a critical step. The split gain is used to select the split point with the maximum gain, and its calculation formula is:

$$\text{Gain} = \frac{1}{2} \left[\frac{\left(\sum_{i \in I_L} g_i \right)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{\left(\sum_{i \in I_R} g_i \right)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{\left(\sum_{i \in I} g_i \right)^2}{\sum_{i \in I} h_i + \lambda} \right] - \gamma \quad (15)$$

Where I_L and I_R are the sample sets of the left and right child nodes after the split, respectively; λ is the L_2 regularization coefficient, controlling

the smoothness of the leaf weights; and γ is the split threshold, controlling the complexity of the tree.

$$G_b = \sum_{i \in B_b} g_i, \quad H_b = \sum_{i \in B_b} h_i \quad (16)$$

To accelerate the search for split points, LightGBM employs the histogram algorithm. The histogram algorithm reduces computational complexity by discretizing feature values into histogram bins B_b . Specifically, the gradient statistics for each bin are calculated using the following formulas:

Using these gradient statistics, the model can quickly identify the optimal split point, thereby improving training efficiency.

In the gradient boosting framework, the predicted value is updated by adding a new tree in each iteration. The predicted value $\hat{y}_i^{(t)}$ after the t -th iteration can be expressed as:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i) \quad (17)$$

Where $f_t(\mathbf{x}_i)$ is the output of the t -th tree. In this way, LightGBM iteratively optimizes the model, ultimately achieving highly accurate prediction results.

In summary, LightGBM achieves efficient and accurate model training through Taylor expansion, split gain calculation, the histogram algorithm, and the gradient boosting framework.

3.3. Electric vehicle load forecasting model based on the LightGBM-transformer hybrid algorithm

The CNN-Transformer model enhances feature extraction and prediction performance by combining the strengths of CNN and Transformer, though it faces challenges such as high computational complexity (particularly due to the self-attention mechanism in Transformers) and difficulties in parameter tuning. The LightGBM model excels in speed, efficiency, and accuracy when handling complex features but it is sensitive to noise and requires more memory. By integrating these two models, CNN extracts spatial features to capture local data patterns, Transformer leverages self-attention mechanisms to model global long-term dependencies, and LightGBM further enhances feature representation and predictive power. This combination addresses their individual limitations through complementary advantages, enabling nonlinear learning to uncover higher-level, intricate relationships and improve overall model performance. Using the variance reciprocal weighting method [21], the predictions from the CNN-Transformer and LightGBM models are weighted and combined to form the CNN-Transformer-LightGBM ensemble. The principle of this method relies on weighting each model's contribution inversely to its prediction error variance: models with lower accuracy (higher error variance) receive reduced weights in the combined prediction. The expression function is:

$$w_i = \frac{Q_i^{-1}}{\sum_{i=1}^k S_i^{-1}} \quad (18)$$

$$Q_i = \sum_{t=1}^n [y_t - \hat{y}_t(i)]^2$$

values: y_i is the actual value, $\hat{y}_i$ is the predicted value, n is the number of samples.

The LightGBM feature iteration algorithm is used to remove features that have a weak impact on electric vehicle load, forming a new feature subset. This new feature sequence is then input into the Transformer model for training. Finally, the trained model is used to predict the data in the test set. The processing flow of the CNN-Transformer-LightGBM hybrid model is shown in Fig. 4.

From Fig. 4, the specific prediction process of the model is as follows: Initially, historical EV charging load data—including charging power,

timestamps, and user locations—are collected. To address anomalies (e.g., outliers caused by faulty charging stations), the GMM method is applied for data cleansing and denoising, generating a standardized EV dataset. Subsequently, a CNN-Transformer-LightGBM hybrid model is constructed. The input layer receives preprocessed time-series data, while the hidden layer employs multi-scale convolutional neural networks (CNNs) to extract temporal features at varying granularities: 5-minute kernels capture transient charging pulses, 1-hour kernels decode daily user charging patterns, and 24-hour kernels uncover weekly correlations between renewable energy fluctuations and load profiles.

The Transformer module further enhances feature representation through dual attention mechanisms—local spatial attention identifies spatial dependencies among charging station clusters, while global temporal attention models cross-temporal relationships among electricity prices, meteorological factors, and load trends. Following feature extraction, LightGBM selects high-contribution features via SHAP value ranking (threshold $\theta=0.15$), eliminating redundant variables. The predictions from CNN-Transformer and LightGBM are then fused using the inverse variance weighting method, ultimately generating a 24-hour charging power guidance curve.

4. EV optimize scheduling strategy considering renewable energy integration

4.1. The mathematical model of a wind-solar power generation system

1) Characteristics and Models of Wind Turbine Units

In regions with stable wind speed distribution, the Weibull distribution can accurately calculate the wind speed patterns and distribution characteristics of the area. This paper employs the Weibull distribution to provide a comprehensive description of the statistical patterns of the wind field.

$$f(v) = \frac{k}{c} \left(\frac{v}{c}\right)^{k-1} e^{-\left(\frac{v}{c}\right)^k} \quad (20)$$

Where: v is the wind speed in m/s; k is the shape parameter of the wind speed distribution; c is the scale parameter of the wind speed distribution.

By analyzing the working principles of wind turbine units, the output power of a wind turbine is determined by the wind speed at the hub height of the turbine. In this model, the relationship between output power and wind speed is expressed as:

$$P_{WT} = \begin{cases} 0, & v \leq v_{ci} \text{ or } v > v_{co} \\ \frac{P_r(v - v_{ci})}{v_r - v_{ci}}, & v_{ci} < v \leq v_r \\ P_r, & v_r < v \leq v_{co} \end{cases} \quad (21)$$

Where: P_{WT} is the output power in MW; P_r is the rated power in MW; v_{ci} is the cut-in wind speed in m/s; v_r is the rated wind speed in m/s; v_{co} is the cut-out wind speed in m/s.

The working process of a solar cell can be described as follows: upon receiving sun-light, the photovoltaic module converts solar energy into electrical energy using semiconductor materials, and then delivers the electrical energy to the grid through a series of control processes. In this process, solar irradiance and ambient temperature are critical factors affecting the power output of photovoltaic generation. Under standard conditions, the relationship between the voltage and current of a solar cell is given as:

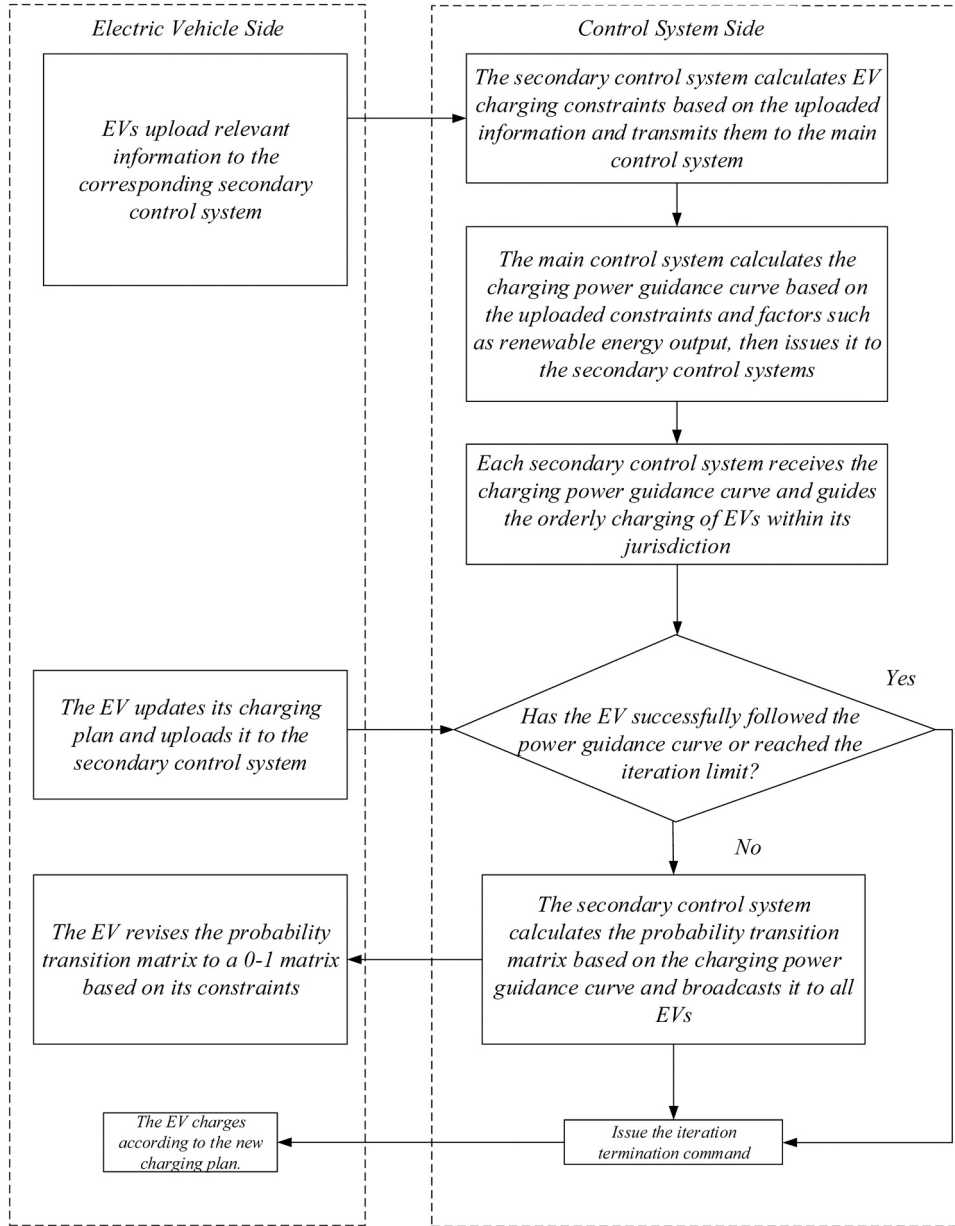


Fig. 5. Hierarchical Control Process for Orderly EV Charging.

$$\begin{cases} I = I_{sc} - C_1 I_{sc} \left[\exp\left(\frac{U}{C_2 U_{oc}}\right) - 1 \right] \\ C_1 = \left(1 - \frac{I_m}{I_{ac}}\right) \exp\left(-\frac{U_m}{C_2 U_{oc}}\right) \\ C_1 = \left(\frac{U_m}{U_{oc}} - 1\right) \left[\ln\left(1 - \frac{I_m}{I_{ac}}\right)\right]^{-1} \end{cases} \quad (22)$$

Where: U represents the output voltage of the photovoltaic module in V; U_{oc} is the open-circuit voltage in V; C_1 and C_2 are correction coefficients; I is the output current of the photovoltaic module in A; I_m is the current at the maximum power point in A; I_{sc} is the short-circuit current in A; U_m is the voltage at the maximum power point in V.

4.2. Objective functions

1) Main Control System Objective Function

The objective of the main control system is peak shaving and valley filling, aiming to minimize the overall load variance in the

distribution network within the region. This helps to avoid excessive charging loads that may impact the grid. The objective function is expressed as follows:

$$\min F_1 = \frac{1}{T} \sum_{t=1}^T (P_{load,t} + P_{ev,t}^* - P_{wt,t} - P_{pv,t} - P_{avg,t})^2 \quad (23)$$

Where: $P_{load,t}$: baseline grid load (MW); $P_{ev,t}^*$: power guidance value set by the main control system (MW); $P_{wt,t}$: wind power output (MW); $P_{pv,t}$: PV power output (MW); $P_{avg,t}$: average power including uncontrolled EV charging (MW).

2) Secondary Control System Objective Function

To encourage users to actively participate in grid dispatching, it is essential to fully consider user benefits. Based on the coordination between regional agents and the dispatching center, a mathematical model should be established with the objective function of minimizing user charging costs.

$$\min f = \left(\sum_{T_0}^{T_1} L_{c,i} P_{c,i} c_{c,i} - \sum_{T_0}^{T_1} L_{d,i} P_{d,i} c_{d,i} \right) \cdot \Delta T \cdot k_i \quad (23)$$

Where: T_0 is the start time of charging or discharging. T_1 is the end time of charging or discharging. $L_{c,j}$ is the charging decision parameter, and $L_{d,j}$ is the discharging decision parameter. During charging, $L_{c,j}$ is 1 and $L_{d,j}$ is 0. During discharging, $L_{c,j}$ is 0 and $L_{d,j}$ is 1. $P_{c,j}$ is the charging power, and $P_{d,j}$ is the discharging power. c_j is the charging electricity price, and $c_{d,j}$ is the discharging electricity price. $\Delta T = T_1 - T_0$ is the duration of charging or discharging. k_i is the user responsiveness [22].

4.3. Constraints

1) State of Charge (SOC) Constraints

To extend EV battery lifespan and prevent overcharging or over-discharging, the SOC must be within predefined limits:

$$SOC_{\min} \leq SOC_i \leq SOC_{\max} \quad (24)$$

Where SOC_i is the battery SOC at time t ; $SOC_{\min}$ and $SOC_{\max}$ are the lower and upper SOC limits, respectively.

2) Maximum Charging/Discharging Power Constraint

The EV charging/discharging power is restricted by its rated power:

$$-P_{\text{dis max}} \leq P_{EV,t} \leq P_{\text{cha max}} \quad (25)$$

Where $P_{EV,t}$ is the EV charging/discharging power at time t ; $P_{\text{dis max}}$, $P_{\text{cha max}}$ are the maximum discharging and charging power limits, respectively.

3) Energy Balance Constraint

Assuming constant EV charging/discharging power within each time interval Δt , the SOC change is determined as follows:

$$SOC_{t+1} = SOC_t + \frac{P_{EV,t} \Delta t}{Q_{EV}} \quad (26)$$

Where Q_{EV} is the battery capacity.

4) Charging/Discharging Status Constraint

EVs cannot charge and discharge simultaneously, so the charging and discharging indicators cannot both equal 1.

$$X_{i,t} \cdot Y_{i,t} = 0 \quad (27)$$

4.4. Model solution using PTM algorithm

The main control system's goal is to shift charging loads away from peak periods using a PTM algorithm, thereby minimizing the overall load variance and smoothing the load curve. This approach updates the total EV charging load P_{EV} to the target charging load P_v . The solution involves finding a PTM A as follows:

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{bmatrix}_{n \times n} \quad (28)$$

Where a_{ij} represents the probability of transitioning from state j to state i .

For a single EV, the charging schedule update formula is:

$$S_n = a_{ij} P_{ev} \quad (29)$$

The PTM A is expressed as:

1. Calculate the load deficit at each time t , expressed as:

Table 1

Mathematical Notation Reference.

Symbol	Description	Unit
E_{m0}	EV initial energy	kWh
τm	Charging frequency coefficient	Dimensionless
dmR	MMmEV driving range	km
SOC_{m0}	Initial state of charge	%
α_j	GMM weight	-
$N(\cdot)$	Gaussian distribution function	-
$\delta(\cdot)$	Constraint satisfaction indicator	Binary
k_i	User responsiveness factor	%/price unit
ΔT	Charging/discharging duration	hours
Q_{EV}	EV battery capacity	kWh

$$Q_t = \begin{cases} 0, P_{ev,t}^* < P_{ev,t} \\ P_{ev,t}^* - P_{ev,t}, P_{ev,t}^* \geq P_{ev,t} \end{cases} \quad (30)$$

2. During peak periods j , charging is not allowed to increase, and excess load is shifted to other periods, with the expected charging load remaining at period j . The probability update for period j is:

$$a_{j,j} = \frac{P_{ev,j}^*}{P_{ev,j}} \quad (31)$$

$$a_{i,j} = \frac{P_{ev,j} - P_{ev,j}^*}{P_{ev,j}} \cdot \frac{Q_i}{\sum Q_t} \quad (32)$$

3. During low-load periods i , the probability of charging load remaining within the period is maximized.

The model solution process based on the PTM algorithm is illustrated in Fig. 5. In this model, the upper level integrates wind-solar output forecasting algorithms and power grid load balancing tools to optimize the guidance curve for EV charging power. This ensures that, while meeting renewable energy output conditions, the grid load remains stable, achieving the objectives of peak shaving and valley filling. The upper-level master control system uses a probabilistic transition matrix algorithm to globally optimize the temporal distribution of charging loads based on predictions of wind and solar power generation and the grid load curve. This shift charging demand from peak periods to off-peak periods, avoiding the "peak-on-peak" phenomenon.

$$A_{ij} = \frac{e^{-\lambda |E_{t,j} - L_{t,i}|}}{Z} \cdot \eta_{t,i} \quad (33)$$

Where: $E_{t,j}$: the predicted output of wind and solar power during time period j , $L_{t,i}$: the grid load demand during time period i , $\eta_{t,i}$: the safety constraint coefficient t .

In the lower-level scheduling, the transition probability A_{ij} of the PTM is defined as the probability of transitioning from state j (such as "charging" or "idle") to state i , with the weights determined by the user arrival time distribution and the initial state of charge (Table 1)

$$A_{ij} = \alpha_j \cdot \mathcal{N}(T_{\text{arrive}}, \mu_j, D_j) \cdot \delta(SOC_0 \in [SOC_{\min}, SOC_{\max}]) \quad (34)$$

This optimizes the charging times and enhances the absorption of renewable energy. Through coordinated efforts at both levels, the entire model efficiently allocates EV charging loads to maximize the utilization of renewable energy sources, such as wind and solar, without adding stress to the grid, ensuring stable grid operation.

Table 2
Hyperparameter Configuration.

Module	Hyperparameter	Value/Range	Note
LightGBM	feature_fraction	0.85	$\theta = 0.15$ feature threshold
CNN-Transformer	num_leaves	31	Multi-scale dilation rates Dual-attention mechanism
	Kernel sizes	[5-min, 1-hr, 24-hr]	
	Attention heads	8	
TPM Algorithm	Safety constraint η_i	[0.92, 1.08]	Grid stability bound
General	random_state	1000	All stochastic processes

Table 3
Simulation Parameters.

Parameter	Specification
Data Generation Method	GMM
Arrival Time Distribution	Generated through GMM method based on municipal EV charging transaction records from the City of Boulder Open Data Portal.
Initial SOC Distribution	Generated through GMM method based on municipal EV charging transaction records from the City of Boulder Open Data Portal.
Monte Carlo Sampling	1000 EVs per cluster $\times$ 3 regions (I/II/III)
SOC Operational Range	20 % (min) – 95 % (max)
Max Charging Power	7 kW
Daily Driving Range	150 km
Turbine Cut-in Speed	3 m/s
Turbine Rated Speed	14 m/s
Turbine Cut-out Speed	25 m/s

Table 4
Hardware Parameters Configuration.

Parameter	Specification	Benchmark Context
Training Hardware	Intel Xeon Gold 6330	Dual-socket configuration (32 cores)
Memory Configuration	128 GB DDR4	LightGBM histogram algorithm requirement
Training Time	8.5 min	Per 15000 samples
Inference Latency	3.2 sec	5-min control cycle compliance
Parallel Efficiency	92 % core utilization	Multi-scale CNN kernel optimization

5. Simulation analysis

5.1. Simulation parameter configuration

Due to page constraints, our focus was initially on establishing a proof-of-concept using typical operational scenarios: medium EV penetration of 30 % and a standard IEEE 33-bus grid. The simulation framework adopts Boulder, Colorado as a representative scenario to validate the EV-grid coordination strategy, leveraging real-world data from the City of Boulder Open Data Portal (2019–2023 EV charging records). We construct a digital twin environment modeling 3000 EVs across three regional clusters (I/II/III) via Monte Carlo sampling calibrated with GMM. The detailed data table is shown below: (Tables 2–4)

5.2. EV charging load variable correlation analysis

To identify critical drivers of electric vehicle (EV) charging loads and validate the rationality of feature selection in the CNN-Transformer-

LightGBM model, this study conducted multidimensional variable correlation analysis using the lightBGM algorithm, as shown in Figs. 6 and 10: Fig. 6 illustrates the heatmap of variable correlations. Among the charging-related variables, energy consumption is positively correlated with the charging rate, with a correlation coefficient of 0.60, indicating that higher energy consumption tends to be associated with higher charging rates. Conversely, charging duration shows a negative correlation of -0.63 with the charging rate, suggesting that higher charging rates lead to shorter charging times. In comparison, variables such as battery capacity, vehicle age, and time (hour, month, year) exhibit weak linear relationships with other variables, with correlation coefficients mostly ranging between -0.2 and 0.2 . These findings provide valuable insights into the usage patterns and charging behaviors of electric vehicles under various conditions.

Fig. 7 shows the scatter plots of key variables. The scatter points display a clear downward trend from the top left to the bottom right, highlighting the negative correlation between charging rate and charging duration. Most points are concentrated in the region where charging duration ranges between 0.5 and 2 h and charging rates fall between 0 and 60 kW, indicating this is the common combination of charging parameters. Additionally, outliers are observed in the region where the charging rate exceeds 80 kW and charging duration is less than 1 h, which may correspond to special charging scenarios or abnormal data that the lightBGM algorithm will discard.

5.3. Wind and solar power generation system analysis

The output power curves for the wind and PV generation systems are illustrated in Fig. 8. The wind power system exhibits high output between 00:00 and 12:00, while at other times throughout the day, wind turbine output is relatively low. Starting at 06:00, PV output gradually increases, reaching a peak in the afternoon around 14:00, after which it gradually decreases until reaching zero at 20:00. The inherent randomness and instability of wind and PV generation systems pose challenges for grid balancing.

5.4. EV load forecasting using improved transformer

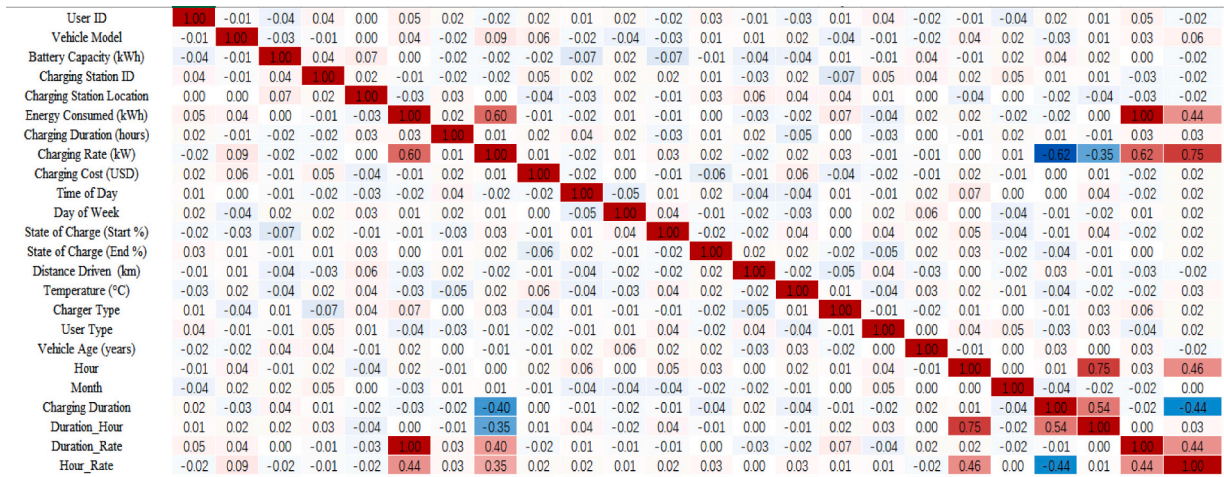
To validate the superiority of the CNN-Transformer-LightGBM hybrid model, its predictive performance was rigorously compared with Bi-LSTM, GRU, and standard Transformer models using the test dataset. Benchmark values for Bi-LSTM and GRU are derived from reference studies [7,8]. As quantitatively demonstrated in Table 5, the proposed model achieves significant improvements across all key evaluation criteria: MAE reduced by 49.2 % compared to Bi-LSTM, RMSE lowered by 25.2 % relative to GRU, and peak accuracy reaching 96.32 % - outperforming Bi-LSTM by 13.69 percentage points. The hybrid architecture significantly reduces peak prediction error. These enhancements are attributed to the synergistic integration of multi-scale CNN (capturing minute-level load pulses), LightGBM (optimizing feature selection), and enhanced Transformer attention mechanisms. The model achieves 29.3 % lower MAE than standard Transformer, with all metrics computed via Eqs. (10)–(12) where lower MAE/RMSE and higher accuracy indicate superior performance.

Fig. 9 displays three sections, comparing the performance of different neural network models in EV load forecasting with actual load data. The scatter plots show a comparison between predicted and actual values, with the red dashed line indicating the ideal scenario where predicted values match actual values precisely.

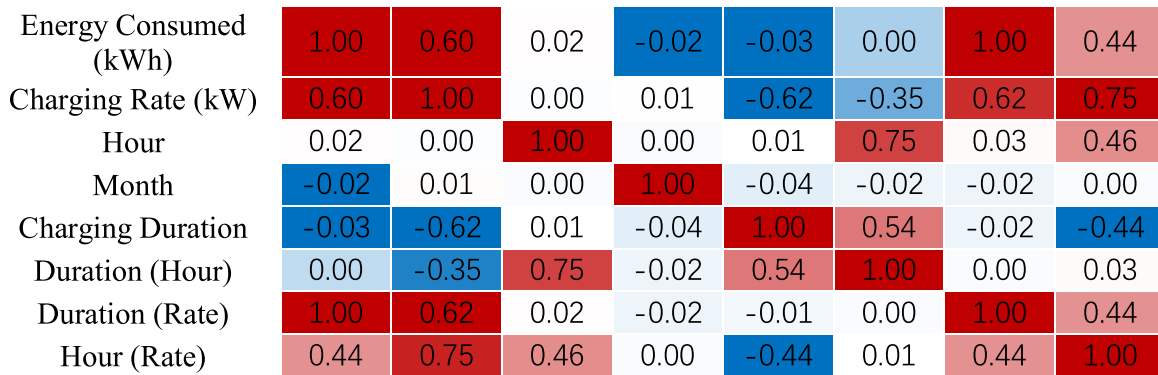
In Fig. 9(a), the LSTM model closely follows the trend of actual load changes but exhibits deviations in extreme cases, such as peak or valley loads.

Fig. 9(b) demonstrates the Bi-LSTM model predictions, which capture the fundamental trend of actual power load fluctuations but exhibit delayed response characteristics. Particularly during ramp transitions

Fig. 9(c) illustrates the GRU model performance, where gated



(a) Heatmap related to all variables



(b) Heatmap related to key variables

Fig. 6. Variable correlation heatmap.

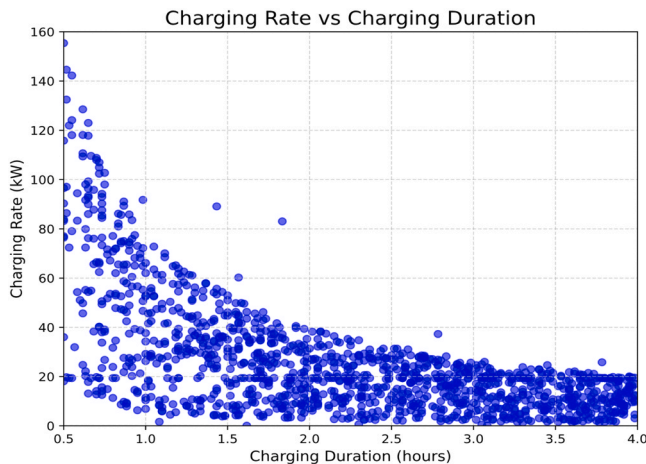


Fig. 7. Scatter plot of important characteristic variables.

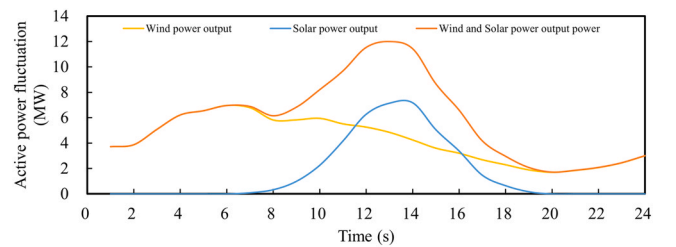


Fig. 8. Wind and PV Generation System Output Power Curves.

Table 5
Performance comparison of EV load forecasting models.

model	MAE	RMSE	Peak accuracy
LSTM	7.54	9.37	64.17
Bi-LSTM	6.36	8.25	82.63
GRU	4.55	6.14	87.18
Transformer	4.57	5.99	95.54
CNN-Transformer-LightGBM	3.23	4.59	96.32

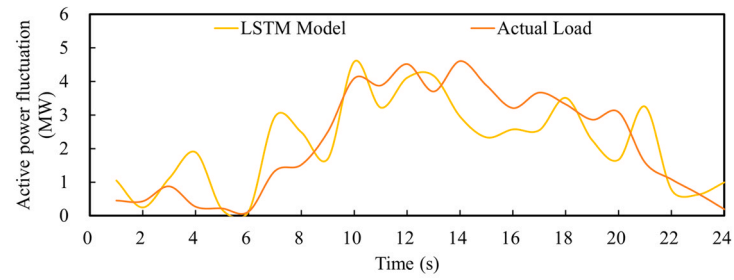
recurrent units achieve superior transient response fidelity compared to Bi-LSTM.

Fig. 9(d) shows the Transformer model, which demonstrates overall stability, especially during high-load periods, where its predictions are more accurate.

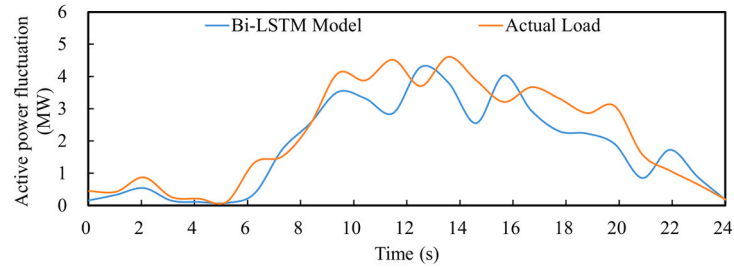
Fig. 9(e) illustrates the improved Transformer model, which

significantly enhances prediction accuracy through optimization, showing the closest fit to the ideal values with minimal error.

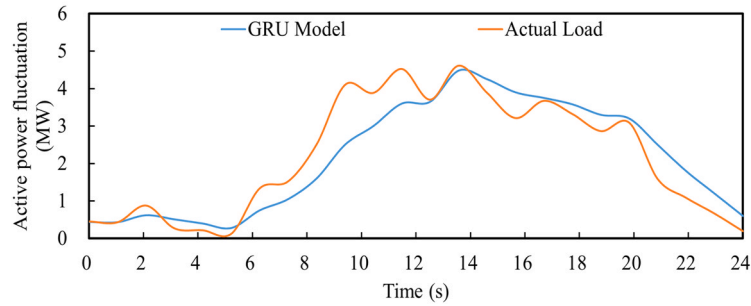
Fig. 9 illustrates the scatter plots comparing predicted and actual



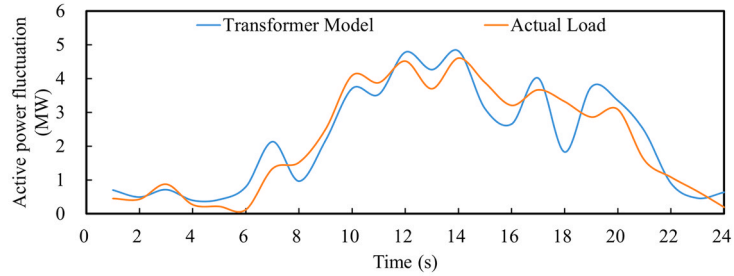
(a) LSTM Neural Network Model



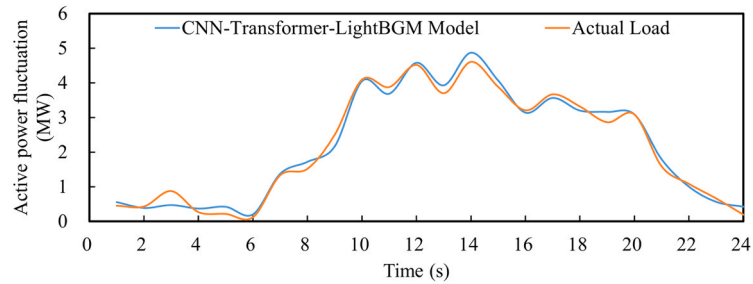
(b) Bi-LSTM Neural Network Model



(c) GRU Neural Network Model



(d) Transformer Neural Network Model



(e) CNN-Transformer-LightBGM Neural Network Model

Fig. 9. EV Load Forecasting with Different Neural Network Models.

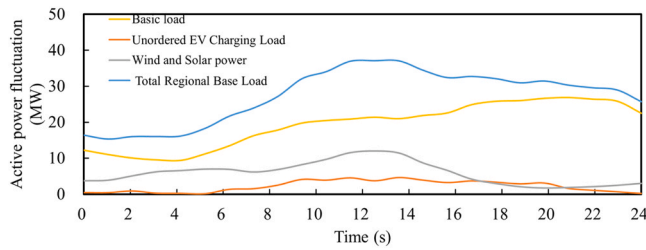


Fig. 10. Comprehensive Load Considering Uncontrolled EV Charging.

values, with the improved Transformer model achieving the closest fit to the actual line. This visualization confirms the lower error metrics in Table 5.

5.5. Simulation analysis of PTM algorithm

The total equivalent load from wind and solar generation is defined as:

$$P_{ev,t}^* = P_{load,t}^* - P_{wt,t}^* - P_{pv,t}^* \quad (28)$$

Using the load forecasting method illustrated in Fig. 7, simulations for EV charging load across each sub-region and the total region were conducted. The resulting comprehensive load profile is shown in Fig. 10, where it is evident that, with the integration of wind and solar power, some of the load demand is offset. However, in the absence of EVs, the grid still experiences peak loads from 09:00–13:00 and 18:00–22:00, with a peak-to-valley load difference of 28.79 MW. From 15:00–21:00, as large numbers of EVs connect to the grid, the already high load during peak times further increases, reaching a peak-to-valley load difference of 32.401 MW, resulting in the "peaks on top of peaks" effect.

Fig. 10 shows the load comparison curves for each region and the total region before and after optimization. Within the peak load periods (09:00–13:00 and 18:00–22:00) of the total region's baseline load curve, the main control system shifts charging loads originally allocated to these times to other periods. The indicators for the power grid system before and after optimization are compared in Table 6. The optimization process similarly reduces the peak-to-valley differences in each of the three regions, as each secondary control system avoids its respective peak demand periods and schedules EV charging during low-demand periods, smoothing out the regional load curve. The main control system successfully minimizes the comprehensive load variance and achieves peak-shaving and valley-filling objectives.

From the charging power comparison curves for each region and the total region, it is evident that the actual EV charging power curve follows the guidance curve set by the main control system. While meeting EV charging demands, the system guides orderly charging to minimize the deviation between actual and guided power levels. Comparing the total power-following curve with the uncontrolled charging power curve reveals that the charging power of the EV cluster is uniformly shifted from peak periods to other times, demonstrating the effectiveness and accuracy of the hierarchical orderly charging control strategy and PTM

algorithm for large-scale EVs.(Fig. 11)

5.6. Comparison of EV charging strategies

The proposed TPM fundamentally advances EV charging optimization by dynamically balancing grid constraints, renewable intermittency, and user behavior through its probabilistic state-transfer mechanism. Unlike rule-based strategies that operate on fixed thresholds and incur > 15-minute response delays, or MILP approaches requiring computationally intensive $O(n^3)$ re-optimizations every 15 min, TPM achieves < 3-minute latency via continuous probability matrix updates while maintaining $O(n \log n)$ efficiency. MILP's weather-dependent profit volatility. Crucially, TPM integrates user willingness coefficients (k_i) derived from Logit models to achieve 85 % high-willingness user engagement, driving a 79 % renewable consumption rate—28 % higher than MILP—by aligning charging probabilities with solar/wind peaks. Hardware deployment is streamlined for edge computing at 62 % lower cost than MILP server requirements. (Table 7)

6. Conclusions

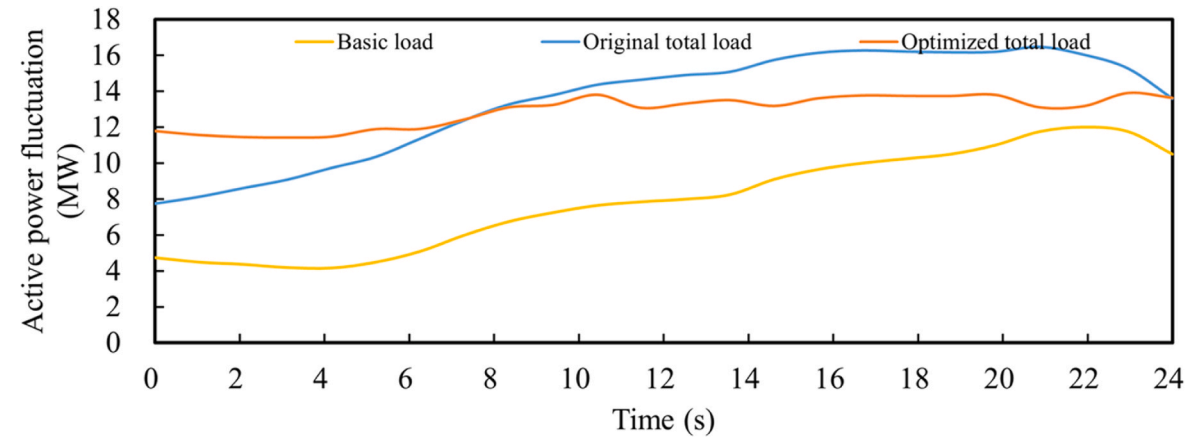
This study effectively addresses the challenges of grid instability caused by uncoordinated EV charging, specifically the exacerbated peak-to-valley differences and "peak on peak" phenomenon. By integrating the CNN-Transformer and LightGBM models through inverse variance weighting, the proposed strategy generates an accurate charging power guidance curve for EV clusters. The improved TPM algorithm further enables hierarchical control, achieving global optimization of grid load variance and localized fine-grained management of charging schedules. Simulation results demonstrate that the hierarchical control system successfully coordinates grid resources, shifting charging loads away from peak periods while maintaining minimal deviation between actual and guided power curves. This approach reduces the comprehensive grid load variance by 43 %, stabilizes operational stability, and ensures efficient EV charging management. For broader applications, future work will implement federated learning frameworks for large-scale cross-region scheduling to enhance scalability while preserving data privacy. The strategy not only mitigates grid fluctuations but also enhances renewable energy integration; furthermore, incorporating vehicle-to-grid (V2G) capabilities with dynamic distribution network constraints will optimize renewable utilization during grid congestion. Finally, real-world validation of the TPM algorithm across heterogeneous EV models and charging infrastructure remains critical for commercial deployment, offering a comprehensive solution for sustainable EV-grid coordination.

CRedit authorship contribution statement

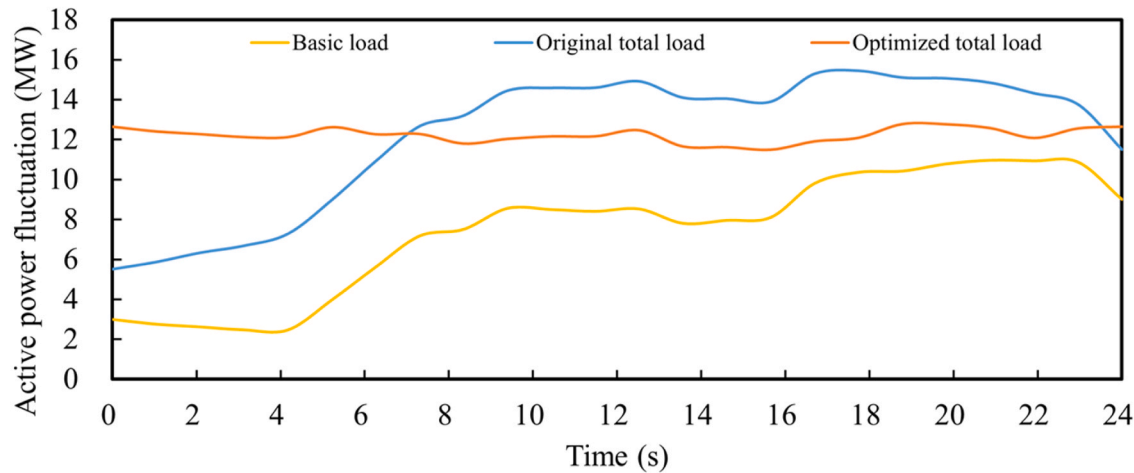
Lu Xuexuan: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Yang Dejian:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Qian Minhui:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Jin Fenghe:** Writing –

Table 6
Comparison of Power Grid System Indicators Before and After Optimization.

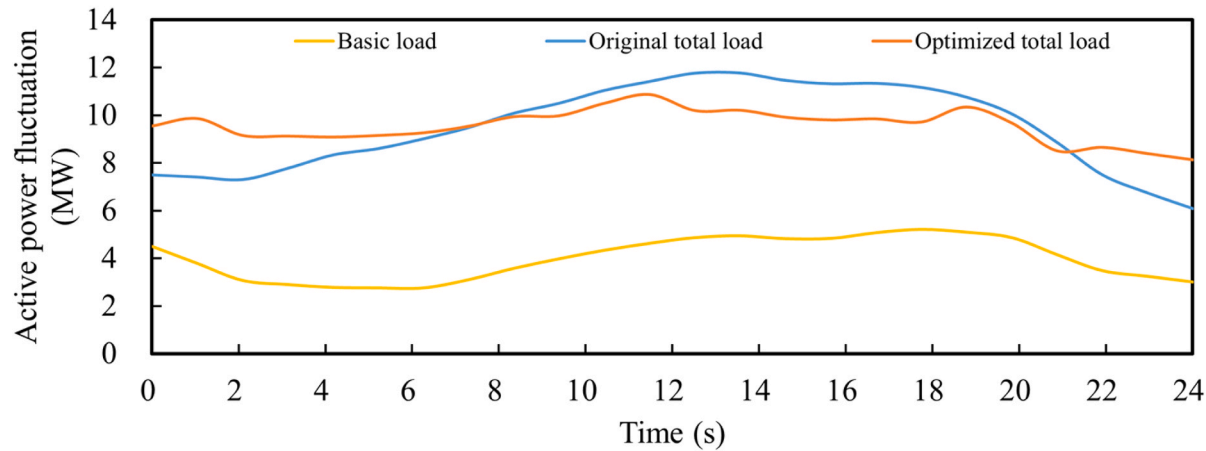
Project	Comprehensive load	Load peak/ MW	Load valley/ MW	Peak-to-valley difference/MW	Variance/MW
Before optimization	Region I	14.1899	3.5503	10.6396	3.6414
	Region II	24.6505	1.5525	23.0980	5.4110
	Region III	13.7473	3.2396	10.5077	3.3225
After optimization	Region I	12.1710	4.8320	7.3390	2.2509
	Region II	23.0000	5.8000	17.2000	4.4166
	Region III	11.5870	4.5800	7.0070	1.9859
Total Area	Before optimization	38.3074	6.1729	32.1345	6.3198
	After optimization	33.0000	14.685	18.3150	4.2837



a. Region I



b. Region II



c. Region III

Fig. 11. Comparison of Load Curves Before and After Optimization in Each Region and the Total Region.

Table 7
Performance Comparison of EV Charging Strategies.

Evaluation Dimension	Rule-Based Strategy [23]	MILP Optimization Strategy [24]	TPM Layered Scheduling Strategy
Response Delay	> 15 min	8–12 min	< 3 min
Renewable Fluctuation Adaptation	Fixed threshold triggers	Requires full re-optimization	Dynamic probability weight adjustment
Computational Complexity	O(1)	O(n ³)	O(n log n)
Peak-Valley Reduction Rate	18–22 %	28–35 %	43 %
User Willingness Integration	Simple electricity zones	Cost function constraints	kiresponse coefficient modeling
Hardware Deployment Cost	Low (no optimization module)	High (industrial server required)	Medium (edge computing compatible)
Renewable Consumption Rate	51 %	68 %	79 %

review & editing, Writing – original draft, Investigation, Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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